

The $C^{1,\alpha}$ Wasserstein–Hilbert lift equivalence fails for every $0 < \alpha < 1$

Candidate full counterexample to Remark 3.12(i) of arXiv:2004.01660

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Status: candidate full counterexample, likely valid. For every $0 < \alpha < 1$, there is a continuously differentiable $\mathcal{U}_\alpha : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}$ whose rearrangement-invariant lift $\tilde{\mathcal{U}}_\alpha : L^2(\Omega) \rightarrow \mathbb{R}$ is globally $C^{1,\alpha}$, while $\mathcal{U}_\alpha$ is not $C^{1,\alpha}$ in Definition 3.8 of Gangbo–Mészáros. The example is the $(1 + \alpha)$ -power of the second-moment radius. It even satisfies the definition’s optimal-transport Taylor estimate; only the required pointwise Hölder regularity on a measure’s support fails. Thus the proposed equivalence is false.

1 The exact open problem

Let Ω be an atomless probability space and $\mathbb{H} = L^2(\Omega; \mathbb{R})$. The source lifts a functional $\mathcal{U} : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$\tilde{\mathcal{U}}(X) = \mathcal{U}(\text{Law}(X)).$$

Its Definition 3.8 says, in particular, that $\mathcal{U} \in C^{1,\alpha}$ only if there is one constant C such that, for every μ ,

$$|\nabla_w \mathcal{U}(\mu)(q) - \nabla_w \mathcal{U}(\mu)(r)| \leq C|q - r|^\alpha \quad (q, r \in \text{supp } \mu). \quad (1)$$

It also requires the first-order remainder along every optimal coupling to be $O(W_2^{1+\alpha})$.

Lemma 3.11 of the source proves equivalence between its Wasserstein $C^{1,1}$ class and the ordinary Hilbert $C^{1,1}$ class. Remark 3.12(i) asks whether the equivalence persists for $C^{1,\alpha}$, $0 < \alpha < 1$ [1]. The complete source pages containing Definition 3.8 and the question are reproduced next.

By this construction, for $t \in [0, 1]$ we have

$$\begin{aligned} U^{(m)}((1-t)q + tq') &= \mathcal{U}(\mu_t^{(m)}) \leq (1-t)\mathcal{U}(\mu_q^{(m)}) + t\mathcal{U}(\mu_{q'}^{(m)}) - \frac{\lambda}{2}t(1-t)W_2(\mu_q^{(m)}, \mu_{q'}^{(m)}) \\ &= (1-t)U^{(m)}(q) + tU^{(m)}(q') - \frac{\lambda}{2m}t(1-t) \sum_{i=1}^m |q_i - q'_i|^2. \end{aligned}$$

Therefore, the $\frac{\lambda}{m}$ -convexity of $U^{(m)}$ in a small neighborhood of q follows.

(3) $\Rightarrow$ (1) We suppose $U^{(m)}$ is $\frac{\lambda}{m}$ -convex for all natural number m . We plan to show the λ -convexity of $\tilde{\mathcal{U}}$ on $\mathbb{H}$. Note the $\frac{\lambda}{m}$ -convexity of $U^{(m)}$ is equivalent to the λ -convexity of the restriction of $\tilde{\mathcal{U}}$ to $\{M^q : q \in \mathbb{R}^{md}\} \subset \mathbb{H}$. In particular, the local Lipschitz constants of these restrictions are bounded from above by a number which is independent of m . These finite dimensional functions then have a unique extension $\tilde{\mathcal{U}}$ on $\mathbb{H}$, which is λ -convex and coincides with $\tilde{\mathcal{U}}$ on a dense subset of $\mathbb{H}$. It suffices to know that $\tilde{\mathcal{U}}$ is continuous to conclude that it is nothing but $\tilde{\mathcal{U}}$. $\square$

3.3. $C^{1,1}$ functions on $(\mathcal{P}_2(\mathbb{M}), W_2)$ versus $C^{1,1}$ functions on $\mathbb{H}$. Given a differentiable function $\mathcal{U} : \mathcal{P}_2(\mathbb{M}) \rightarrow \mathbb{R}$ (cf. [6]), we denote as $\nabla_w \mathcal{U}$ the Wasserstein gradient field of $\mathcal{U}$. This subsection exploits the connection between the differential of $\mathcal{U} : \mathcal{P}_2(\mathbb{M}) \rightarrow \mathbb{R}$ and the differential of its lift $\tilde{\mathcal{U}} : \mathbb{H} \rightarrow \mathbb{R}$ ([32]). More precisely, we have the following result.

Remark 3.7. Let $x \in \mathbb{H}$ and set $\mu := \sharp(x)$. Then $\mathcal{U}$ is differentiable at μ if and only if $\tilde{\mathcal{U}}$ is differentiable at x and in this case, we have the factorization $\nabla \tilde{\mathcal{U}}(x) = \nabla_w \mathcal{U}(\mu) \circ x$.

Definition 3.8. Let $\mathcal{B} \subseteq \mathcal{P}_2(\mathbb{M})$ be open and geodesically convex. Let $\alpha \in (0, 1]$. We say that $\mathcal{U} \in C^{1,\alpha}(\mathcal{B})$, if it is continuously differentiable on $\mathcal{B}$ and there exists a constant $C \geq 0$ such that

- (1) $\text{spt}(\mu) \ni q_1 \mapsto \nabla_w \mathcal{U}(\mu)(q_1)$ is α -Hölder continuous (or simply Lipschitz continuous if $\alpha = 1$) with constant C for any $\mu \in \mathcal{B}$.

$$(2) \left| \mathcal{U}(\nu) - \mathcal{U}(\mu) - \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_2 - q_1) d\gamma(q_1, q_2) \right| \leq CW_2^{1+\alpha}(\mu, \nu), \quad \forall \mu, \nu \in \mathcal{B}, \quad \forall \gamma \in \Gamma_o(\mu, \nu).$$

Definition 3.9. Similarly to the previous definition, let $\mathcal{B} \subseteq \mathcal{P}_2(\mathbb{M})$ be open and geodesically convex and let $K \subseteq \mathbb{M}$ be a convex open set. Let $\alpha \in (0, 1]$. We say that $u \in C^{1,\alpha}(K \times \mathcal{B})$, if it is continuously differentiable on $K \times \mathcal{B}$ and there exists a constant $C \geq 0$ such that

- (1) $\text{spt}(\mu) \ni q_1 \mapsto \nabla_w u(q, \mu)(q_1)$ is α -Hölder continuous (or simply Lipschitz continuous if $\alpha = 1$) with constant C for any $(q, \mu) \in K \times \mathcal{B}$.

(2)

$$\begin{aligned} \left| u(\bar{q}, \nu) - u(q, \mu) - D_q u(q, \mu) \cdot (\bar{q} - q) - \int_{\mathbb{M}^2} \nabla_w u(q, \mu)(q_1) \cdot (q_2 - q_1) d\gamma(q_1, q_2) \right| \\ \leq C (|\bar{q} - q|^{1+\alpha} + W_2^{1+\alpha}(\mu, \nu)), \quad \forall \bar{q}, q \in K, \mu, \nu \in \mathcal{B}, \quad \forall \gamma \in \Gamma_o(\mu, \nu). \end{aligned}$$

Remark 3.10. (i) Let us notice that Definition 3.8(2) implies that $\nabla_w \mathcal{U}$ is ' α -Hölder continuous' in the following sense. We have

$$\left| \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_1 - q_2) d\gamma(q_1, q_2) - \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\nu)(q_2) \cdot (q_1 - q_2) d\tilde{\gamma}(q_2, q_1) \right| \leq 2CW_2^{1+\alpha}(\mu, \nu),$$

for any $\mu, \nu \in \mathcal{B}$ and $\gamma \in \Gamma_o(\mu, \nu)$, $\tilde{\gamma} \in \Gamma_o(\nu, \mu)$.

- (ii) Let us underline that the inequality in Definition 3.8(2) naturally encodes also the fact that $\mathcal{U}$ is locally Lipschitz continuous. Indeed, that inequality, implies that

$$\begin{aligned} |\mathcal{U}(\nu) - \mathcal{U}(\mu)| &\leq CW_2^{1+\alpha}(\mu, \nu) + \int_{\mathbb{M}^2} |\nabla_w \mathcal{U}(\mu)(q_1)| \cdot |q_2 - q_1| d\gamma(q_1, q_2) \\ &\leq CW_2^{1+\alpha}(\mu, \nu) + \|\nabla_w \mathcal{U}(\mu)\|_{L^2(\mu)} W_2(\mu, \nu) = \left(CW_2^\alpha(\mu, \nu) + \|\nabla_w \mathcal{U}(\mu)\|_{L^2(\mu)} \right) W_2(\mu, \nu), \end{aligned}$$

Now, let $\mu, \nu \in \mathcal{P}(\mathbb{M})$ and $x, y \in \mathbb{H}$ such that $\sharp(x) = \mu$, $\sharp(y) = \nu$ and $W_2(\mu, \nu) = \|x - y\|$. Let us note that $\gamma := \sharp(x, y) \in \Gamma_o(\mu, \nu)$. We have

$$\nabla \tilde{\mathcal{U}}(x)(y - x) = \int_{\Omega} \nabla_w \mathcal{U}(\mu)(x(\omega)) \cdot (y(\omega) - x(\omega)) d\omega = \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_2 - q_1) d\gamma(q_1, q_2).$$

Thus, by (3.1)

$$\left| \mathcal{U}(\nu) - \mathcal{U}(\mu) - \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_2 - q_1) d\gamma(q_1, q_2) \right| \leq \frac{C}{2} W_2^2(\mu, \nu),$$

which by the arbitrariness of μ, ν implies the statement.

Part 2. We now need to prove the reversed implication and start by assuming that $\mathcal{U}$ is $C^{1,1}(\mathcal{P}_2(\mathbb{M}))$. In particular $\nabla_w \mathcal{U}(\mu)(\cdot)$ is C -Lipschitz continuous on $\text{spt}(\mu)$ (uniformly in μ) and increasing the value of C if necessary, we assume the inequality in Definition 3.8(2) to hold with the same constant C . Take $x, y \in \mathbb{H}$ and set $\mu := \sharp(x)$ and $\nu := \sharp(y)$. Recall $\tilde{\mathcal{U}} \in C^1(\mathbb{H})$ and $\nabla \tilde{\mathcal{U}}(x) = \nabla_w \mathcal{U}(\mu) \circ x$. Let $\gamma := \sharp(x, y)$ and let $\gamma_0 \in \Gamma_o(\mu, \nu)$. We have

$$\begin{aligned} & \left| \tilde{\mathcal{U}}(y) - \tilde{\mathcal{U}}(x) - \nabla \tilde{\mathcal{U}}(x)(y - x) \right| \\ &= \left| \mathcal{U}(\nu) - \mathcal{U}(\mu) - \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_2 - q_1) d\gamma(q_1, q_2) \right| \\ &\leq \left| \mathcal{U}(\nu) - \mathcal{U}(\mu) - \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_2 - q_1) d\gamma_0(q_1, q_2) \right| \\ &\quad + \left| \int_{\mathbb{M}^2} \nabla_w \mathcal{U}(\mu)(q_1) \cdot (q_2 - q_1) d(\gamma_0 - \gamma)(q_1, q_2) \right| \\ &\leq C W_2^2(\mu, \nu) + \frac{1}{2} \|D_q \nabla_w \mathcal{U}(\mu)\|_{L^\infty} \left(\int_{\mathbb{M}^2} |q_1 - q_2|^2 d\gamma(q_1, q_2) + \int_{\mathbb{M}^2} |q_1 - q_2|^2 d\gamma_0(q_1, q_2) \right) \\ &\leq C W_2^2(\mu, \nu) + \frac{1}{2} C (\|x - y\|^2 + W_2^2(\mu, \nu)) \leq 2C \|x - y\|^2, \end{aligned}$$

where in the penultimate line we used an inequality from Lemma 3.3 [32]. Indeed, according to Lemma 3.3 [32] if $\gamma_1, \gamma_2 \in \Gamma(\mu, \nu)$ and $\xi \in C_c^2(\mathbb{M})$, then

$$\left| \int_{\mathbb{M}^2} D\xi(q_1) \cdot (q_2 - q_1) d(\gamma_1 - \gamma_2)(q_1, q_2) \right| \leq \frac{1}{2} \|D^2 \xi\|_{L^\infty} \left(\int_{\mathbb{M}^2} |q_1 - q_2|^2 d(\gamma_1 + \gamma_2)(q_1, q_2) \right).$$

Since $\nabla_w \mathcal{U}(\mu)$ is the limit of $(D\xi_n)_{n \in \mathbb{N}}$ (where $(\xi_n)_{n \in \mathbb{N}} \in C_c^\infty(\mathbb{M})$) in $L_\mu^2(\mathbb{M}; \mathbb{R}^d)$ and $\nabla_w \mathcal{U}(\mu)$ has a global Lipschitz continuous extension to $\mathbb{M}$, it is easy to see that the previous inequality is still valid for $D\xi = \nabla_w \mathcal{U}(\mu)$ (for which we use its Lipschitz continuous extension to $\mathbb{M}$).

This completes the verification of the proof of the lemma. $\square$

Remark 3.12. (i) It seems an interesting open problem whether the equivalence in Lemma 3.11 hold for $C^{1,\alpha}$ functions for $\alpha \in (0, 1)$.

(ii) The uniform Lipschitz continuity property of $q \mapsto \nabla_w \mathcal{U}(\mu)(q)$, from the proof of Lemma 3.11, appeared already in [15, Lemma 3.3] and in [16, Proposition 5.36]. However, not only our proof is based on a different approach, it is considerably shorter and will be useful in the proof of Lemma A.1.

Definition 3.13. Let $\mathcal{B} \subseteq \mathcal{P}_2(\mathbb{M})$ be open and geodesically convex and let $\alpha \in (0, 1]$. We say that $\mathcal{U} \in C^{2,\alpha,w}(\mathcal{B})$, if $\mathcal{U} \in C^{1,\alpha}(\mathcal{B})$, and if there exist a constant $C > 0$, and functions

$$\Lambda_0 : \mathbb{R}^d \times \mathcal{B} \rightarrow \mathbb{R}^{d \times d}, \quad \Lambda_1 : \mathbb{M}^2 \times \mathcal{B} \rightarrow \mathbb{R}^{d \times d}$$

such that

$$\Lambda_0 \in L^\infty(\mathbb{M}; \mu), \quad \Lambda_1 \in L^\infty(\mathbb{M}^2; \mu \otimes \mu)$$

(1)

$$\left| \nabla_w \mathcal{U}(\nu)(\overline{q}_1) - \nabla_w \mathcal{U}(\mu)(q_1) - \Lambda_0(q_1, \mu)(\overline{q}_1 - q_1) - \int_{\mathbb{M}^2} \Lambda_1(q_1, a, \mu)(b - a) d\gamma(a, b) \right| \leq C (|q_1 - \overline{q}_1|^{1+\alpha} + W_2(\mu, \nu)^{1+\alpha})$$

2 The counterexample

Fix $0 < \alpha < 1$. For $\mu \in \mathcal{P}_2(\mathbb{R})$, write

$$m_2(\mu) = \int_{\mathbb{R}} q^2 d\mu(q)$$

and define

$$\mathcal{U}_\alpha(\mu) = \frac{1}{1+\alpha} m_2(\mu)^{(1+\alpha)/2}. \quad (2)$$

Theorem 1. *The lift of $\mathcal{U}_\alpha$ belongs to $C^{1,\alpha}(\mathbb{H})$, globally. The functional $\mathcal{U}_\alpha$ is continuously Wasserstein differentiable and obeys the $O(W_2^{1+\alpha})$ Taylor-remainder clause in Definition 3.8. Nevertheless, it violates the supportwise Hölder clause (1). Consequently the $C^{1,\alpha}$ equivalence in Remark 3.12(i) is false for every $\alpha \in (0, 1)$.*

The Hilbert estimate behind the example is elementary.

Lemma 2. *On any real Hilbert space H , the map*

$$J_\alpha(z) = \begin{cases} \|z\|^{\alpha-1}z, & z \neq 0, \\ 0, & z = 0, \end{cases}$$

satisfies

$$\|J_\alpha(x) - J_\alpha(y)\| \leq 3\|x - y\|^\alpha \quad (x, y \in H). \quad (3)$$

Proof. Set $r = \|x\|$, $s = \|y\|$, and $d = \|x - y\|$, and suppose $r \geq s > 0$; the cases with a zero vector are immediate. With $\hat{x} = x/r$ and $\hat{y} = y/s$,

$$\|J_\alpha(x) - J_\alpha(y)\| \leq r^\alpha - s^\alpha + s^\alpha \|\hat{x} - \hat{y}\|.$$

Concavity gives $r^\alpha - s^\alpha \leq (r - s)^\alpha \leq d^\alpha$. Also

$$\|\hat{x} - \hat{y}\| \leq \frac{d}{r} + \frac{r - s}{r} \leq \frac{2d}{r}.$$

If $d \leq r$, the second term is at most $2r^{\alpha-1}d \leq 2d^\alpha$. If $d > r$, it is at most $2s^\alpha \leq 2r^\alpha < 2d^\alpha$. This proves (3). $\square$

3 Verification of every hypothesis

If $X \in \mathbb{H}$, then (2) lifts to the radial functional

$$\tilde{\mathcal{U}}_\alpha(X) = \frac{1}{1+\alpha} \|X\|_2^{1+\alpha}. \quad (4)$$

It is Fréchet differentiable everywhere and

$$\nabla \tilde{\mathcal{U}}_\alpha(X) = J_\alpha(X). \quad (5)$$

Lemma 2 therefore proves the asserted global $C^{1,\alpha}$ regularity. It also gives the explicit Taylor estimate

$$\begin{aligned} |\tilde{\mathcal{U}}_\alpha(Y) - \tilde{\mathcal{U}}_\alpha(X) - \langle J_\alpha(X), Y - X \rangle| &\leq \int_0^1 \|J_\alpha(X + t(Y - X)) - J_\alpha(X)\|_2 \|Y - X\|_2 dt \\ &\leq \frac{3}{1+\alpha} \|Y - X\|_2^{1+\alpha}. \end{aligned} \quad (6)$$

Now let X have law μ . The derivative in (5) factors through X , so the Wasserstein gradient is

$$\nabla_w \mathcal{U}_\alpha(\mu)(q) = \begin{cases} m_2(\mu)^{(\alpha-1)/2} q, & m_2(\mu) > 0, \\ 0, & m_2(\mu) = 0. \end{cases} \quad (7)$$

This field is a gradient in q , hence belongs to the Wasserstein tangent space. Its continuity also follows directly: if $\mu_n \rightarrow \mu$ in W_2 , choose optimal lifts $X_n \rightarrow X$ in L^2 ; then (3) gives

$$\|J_\alpha(X_n) - J_\alpha(X)\|_2 \leq 3\|X_n - X\|_2^\alpha \rightarrow 0.$$

Thus $\mathcal{U}_\alpha$ is continuously Wasserstein differentiable, including at δ_0 .

Moreover, for any optimal coupling $\gamma = \text{Law}(X, Y)$, the linear term in (6) is

$$\langle J_\alpha(X), Y - X \rangle = \int_{\mathbb{R}^2} \nabla_w \mathcal{U}_\alpha(\mu)(q)(r - q) d\gamma(q, r),$$

and $\|X - Y\|_2 = W_2(\mu, \nu)$. Hence (6) is precisely the $O(W_2^{1+\alpha})$ clause of Definition 3.8, with constant $3/(1 + \alpha)$.

4 The decisive failure

Take the standard Gaussian measure γ on $\mathbb{R}$. It has $m_2(\gamma) = 1$ and $\text{supp } \gamma = \mathbb{R}$, so (7) becomes

$$\nabla_w \mathcal{U}_\alpha(\gamma)(q) = q.$$

If the supportwise Hölder requirement (1) held with any finite constant C , then for every $t > 0$,

$$t = |\nabla_w \mathcal{U}_\alpha(\gamma)(t) - \nabla_w \mathcal{U}_\alpha(\gamma)(0)| \leq Ct^\alpha,$$

or $t^{1-\alpha} \leq C$, which is impossible as $t \rightarrow \infty$. No change of representative can repair this: a continuous representative agreeing with $q \mapsto q$ Gaussian-almost everywhere must agree with it on all of $\mathbb{R}$.

The same argument rules out a local version on any Wasserstein neighborhood containing γ , because the failure already occurs for the single measure γ . Thus it is not an artifact of asking for a global estimate across distant probability measures. What breaks for $\alpha < 1$ is the passage from an L^2 -Hölder estimate on rearrangements to a pointwise Hölder estimate on an unbounded support. For $\alpha = 1$, the small-set scaling used in the source's proof is exactly balanced; for $\alpha < 1$, it is not.

5 Conclusion

The answer to Remark 3.12(i) is negative for every exponent it asks about. The counterexample is scalar, explicit, rearrangement invariant, and globally $C^{1,\alpha}$ on the Hilbert lift. It passes the Wasserstein differentiability and optimal-coupling Taylor tests, while failing only the additional supportwise regularity built into the source's definition. Exact-phrase and citation searches found the published source still stating the problem and no later resolution of this specific equivalence.

References

- [1] W. Gangbo and A. R. Mészáros, *Global well-posedness of Master equations for deterministic displacement convex potential mean field games*, Comm. Pure Appl. Math. **75** (2022), 2685–2801; arXiv:2004.01660, Remark 3.12(i).